Data Description

Following Fazzari, Morley, and Panovska (2014)[1], I use lagged mean-adjusted capacity utilization as an observable measure of economic slack. As a procyclical indicator, capacity utilization generally rises during expansions and declines during economic downturns. It therefore provides a useful measure for distinguishing between periods of relatively low and high resource utilization.

The other variables used in the analysis are real government spending, real net taxes, and real GDP.

Data sources: I use quarterly data from 1967 Q1 to 2022 Q4. The starting date is determined by the availability of the capacity utilization series. The monthly capacity utilization series is obtained from the Federal Reserve Statistical Releases website and converted to quarterly frequency using the simple arithmetic mean of the monthly observations within each quarter. The remaining variables are obtained from the National Income and Product Accounts (NIPA) of the Bureau of Economic Analysis (BEA).

The columns of interest from the underlying data file are listed below:

- **Columns 1 and 2: Year and quarter.**
- **Column 4: `tgc` (government consumption).** Government consumption expenditures measure the value of goods and services provided by the government, including expenditures on areas such as education, law enforcement, public administration, and national defense.
- **Column 5: `tgi` (government investment).** Government gross investment includes government expenditures on fixed assets, including structures, equipment, and intellectual property products.
- **Total government spending.** Total government spending is defined as the sum of government consumption and government gross investment:

  $$G_t^{\text{nominal}} = \texttt{tgc}_t + \texttt{tgi}_t.$$

  This measure excludes government transfers.
- **Column 6: `gdp` (nominal gross domestic product).**
- **Column 7: `gdpdef` (GDP price deflator).** The GDP price deflator is used to convert nominal government spending, net taxes, and GDP into real quantities.
- **Column 8: `govrev` (nominal net taxes).** Net taxes are defined as government receipts from direct and indirect taxes, net of transfers to businesses and individuals, following Auerbach and Gorodnichenko (2012).

[1]Fazzari, Steven M., Morley, James & Panovska, Irina. "State-dependent effects of fiscal policy" Studies in Nonlinear Dynamics & Econometrics, vol. 19, no. 3, 2015, pp. 285-315.

- **Column 9: `util` (capacity utilization).** The capacity utilization rate is measured as the ratio of a seasonally adjusted output index to an estimate of potential productive capacity.
- **Column 10: `deutil` (mean-adjusted capacity utilization).** This variable is constructed by subtracting the relevant segment-specific mean from the observed capacity utilization rate.

Data Transformation

Identification of structural breaks: The Bai–Perron sequential test identifies two structural breaks in the capacity utilization series, occurring approximately in 1974 Q3 and 2000 Q4. These breaks, highlighted by the vertical dashed lines in Figure 1, divide the sample into three segments with estimated means of approximately 84.80, 81.20, and 77.08 percent, respectively.

The resulting downward staircase pattern indicates a secular decline in average industrial capacity utilization over the sample period, rather than purely cyclical fluctuations around a constant long-run mean. The first break occurs near the oil-price and productivity disruptions of the mid-1970s. The final segment, beginning around 2000, has the lowest mean and contains the deepest troughs in the sample. In particular, both the 2008–09 Great Recession and the COVID-19 pandemic produced sharp declines in capacity utilization, followed by only partial recoveries. This time-varying mean motivates adjusting capacity utilization before conducting the empirical analysis.

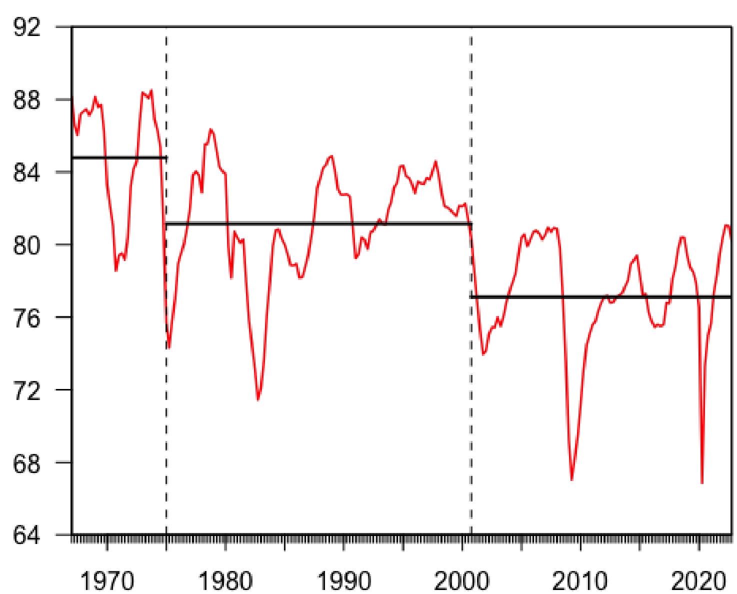

Figure 1: Capacity utilization and estimated structural breaks, 1967 Q1–2022 Q4.

Real government spending, real net taxes, and real GDP enter the VAR in growth rates, while mean-adjusted capacity utilization serves as the threshold variable and enters in levels. The stationarity of the transformed series is assessed using the augmented Dickey–Fuller (ADF) and Kwiatkowski–Phillips–Schmidt–Shin (KPSS) tests, both of which support stationarity.